

Drowning

(Fatal Drowning; Nonfatal Drowning)

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Drowning occurs when submersion in liquid causes suffocation or interferes with breathing.

- During drowning, the body is deprived of oxygen, which can damage organs, particularly the brain.
- Doctors evaluate people for oxygen deprivation and problems that often accompany drowning (such as spinal injuries caused by diving).
- Treatment focuses on correcting oxygen deprivation and other problems.

Drowning may be nonfatal (previously described as near drowning) or fatal. About 4 times as many people are hospitalized for nonfatal drowning as die as a result of drowning.

Drowning is among the top 10 causes of accidental death around the world. In the United States in 2023, drowning was the leading cause of injury-related death in children aged 1 to 4 years, and it was second only to motor vehicle collisions as a cause of death among children aged 5 to 14 years. Drowning was also among the top 10 causes of unintentional death in people under 55 years of age.

Specific groups at higher risk of drowning death include the following:

- People who have taken alcohol or other drugs that affect judgment and alertness
- People who have conditions that cause temporary incapacitation, such as [seizures](#)
- People with [long QT syndrome](#) and certain other disorders that cause abnormal heart rhythms ([arrhythmias](#))
- Children from African American or Native American families

- Children who have not received formal swimming lessons and are unsupervised around water
- Males
- People who engage in dangerous underwater breath-holding behaviors (DUBBs)

More than half of adult drowning deaths involve alcohol. The higher the blood level alcohol level, the greater the risk of a fatal drowning.

Seizure disorders as well as disorders that cause abnormal heart rhythms increase the risk of drowning deaths.

Males are at particular risk of drowning. In fact, 80% of victims over age 1 year are male.

Settings and activities that increase risk of drowning

Drowning is common in pools, hot tubs, and natural water settings. Children and toddlers are also at risk even around small amounts of water, such as in toilets, bathtubs, and buckets of water or other fluids, because they may be unable to escape after falling in.

Diving, particularly into shallow water, may cause [spinal injuries](#) or [head injuries](#) that increase the chance of drowning.

Dangerous underwater breath-holding behaviors (DUBBs) are practiced mostly by healthy young men (often good swimmers) trying to prolong their capacity to remain submerged. There are 3 described types of DUBB:

- Intentional hyperventilation—rapid breathing before submerging lowers carbon dioxide levels, prolonging the time before the levels become high enough to signal the need to return to the surface and breathe
- Hypoxic training—breath-holding while in motion to increase capacity for underwater distance swimming. Athletes who practice hypoxic training should do so only under strict supervision of rescuers informed of the athletes' training intention.
- Static apnea—breath-holding for as long as possible while submerged and motionless, including as a game

In DUBBs, people who intentionally hold their breath under water for extended periods may pass out (called hypoxic blackout or breath-hold blackout) and sometimes drown.

Did You Know...

- Hyperventilating before swimming underwater in an attempt to increase breath-holding time can increase the risk of drowning.

Oxygen deprivation due to drowning

When people are submerged under water, 1 of 2 things can happen:

- Water enters the lungs.
- The vocal cords may go into severe spasm, temporarily preventing water from reaching the lungs but also preventing breathing.

In either case, the lungs cannot transfer oxygen to the blood. The decrease in the level of oxygen in the blood that results may lead to brain damage and death.

Water in the lungs, in very large amounts, causes drowning immediately. Smaller amounts, particularly water that is contaminated by bacteria, algae, sand, dirt, chemicals, or vomit, can cause lung injury that is not apparent until hours after the person has been removed from the water. This problem is sometimes called secondary drowning. Lung injury tends to cause continuing oxygen deprivation. Freshwater in the lungs is absorbed into the bloodstream.

Vocal cord spasm may develop only after a person is removed from the water. If so, it usually occurs within a few minutes. Because water does not enter the lungs, this condition is sometimes called dry drowning.

Effects of submersion in cold water

Submersion in cold water has both good and bad effects. Cooling of the muscles makes swimming difficult, and dangerously low body temperature ([hypothermia](#)) can impair judgment. Cold, however, protects tissues from the ill effects of oxygen deprivation. In addition, cold water may stimulate the mammalian diving reflex, which may prolong survival in cold water. The diving reflex slows the heartbeat and redirects the flow of blood from the hands, feet, and intestine to the heart and brain, thus helping to preserve these vital organs. The diving reflex is more pronounced in children than in adults, so children have a greater chance of surviving prolonged submersion in cold water than do adults.

Did You Know...

- Children are more likely than adults to survive after prolonged submersion.

Symptoms of Drowning

People who are drowning and struggling to breathe are usually unable to call for help. Children who are unable to swim may submerge in less than 1 minute. Adults may struggle longer.

People who are rescued may have a wide range of symptoms and findings. Some are only mildly anxious, whereas others are near death. They may be alert, drowsy, or unconscious. Some people may not be breathing. People who are breathing may gasp for breath or vomit, cough, or wheeze. The skin may appear blue ([cyanosis](#)) indicating insufficient oxygen in the blood. In some cases, respiratory problems may not become evident for several hours after submersion.

Complications of drowning

Some people who are revived after prolonged submersion have permanent brain damage because of the lack of oxygen. People who inhale foreign particles may develop secondary drowning, with [aspiration pneumonia](#) or [acute respiratory distress syndrome](#), causing prolonged difficulty breathing. Such difficulty breathing may not become severe or even apparent until hours after being removed from the water. People who drown in cold water often have [hypothermia](#).

Diagnosis of Drowning

- Doctor's evaluation of symptoms
- Measurement of oxygen in the blood

Doctors diagnose drowning based on the events and the symptoms. Measurement of the level of oxygen in the blood and chest x-rays help reveal the extent of lung damage. Body temperature is measured to check for hypothermia.

Other tests, such as x-rays and computed tomography (CT), may be done to diagnose [head injuries](#) or [spinal injuries](#) in people who may have been injured in a diving

accident. An electrocardiogram (ECG) and sometimes blood tests may be done to diagnose disorders that may have contributed to drowning. For example, certain previously unrecognized [heart arrhythmias](#) can cause unconsciousness while swimming.

Treatment of Drowning

- Artificial respiration and cardiopulmonary resuscitation (CPR)
- Oxygen

Out of the hospital

Immediate on-site resuscitation is the key to increasing the chance of survival without brain damage. Attempts should be made to revive people even when the time under water is prolonged. [Artificial respiration and CPR](#) should be provided as necessary. Rescue breathing is begun before chest compressions, unlike in most other conditions in which CPR is done.

The neck should be moved as little as possible if there is a chance of spinal injury. People who were submerged involuntarily or have any symptoms must be transported to a hospital, by ambulance if possible. People who were submerged but have only mild symptoms may be discharged to their home after several hours of observation in the emergency department. If symptoms persist for a few hours, or if the level of oxygen in the blood is low, people need to be admitted to the hospital.

In the hospital

Most people need supplemental oxygen, sometimes in high concentrations or given via a ventilator at high pressures. If wheezing develops, bronchodilators can help. If an infection develops, antibiotics are given.

If the water was cold, people may have a dangerously low body temperature ([hypothermia](#)) and may need warming. [Spinal injury](#) requires special treatment.

Prognosis for Drowning

The factors that most increase the chances of survival without permanent brain and lung damage are the following:

- Beginning resuscitation immediately (most important)

- Brief duration of submersion
- Cold water temperature
- Young age

Some children have survived without permanent brain damage after submersion for as long as 60 minutes in cold water. Many people who need cardiopulmonary resuscitation can also recover fully, and almost all people who are alert and conscious upon their arrival at the hospital recover fully. People who have consumed alcoholic beverages before submersion are more likely to die or develop brain or lung damage.

Prevention of Drowning

People should not use alcohol or drugs before or during swimming, while boating (even as a passenger), or when supervising children around water.

Water safety and children

Swimming pools should comply with local laws regarding pool safety, including being adequately fenced, because they are one of the most common sites of drowning accidents. In addition, all doors and gates leading to the pool area should be locked. Children in or near any body of water, including pools and bathtubs, need constant supervision, regardless of whether flotation devices are used. Ideally, supervision should be at arm's length. Because infants and young children can drown in only a few inches of water, even water-filled containers, such as buckets or ice chests, are hazardous. Adults should remove water from these containers immediately after use.

Small children should wear approved life jackets or personal flotation devices when playing near bodies of water. Air-filled swimming aids and foam toys (water wings, noodles, and similar items) are not designed to keep swimmers safe and should not be used as substitutes for approved equipment.

Formal swimming lessons reduce the risk of fatal drowning in children aged 1 to 4 years. Swimming lessons are a good idea for all children. However, even children who have taken swimming lessons should be supervised when they are in or around water.

Swimming safety

Swimmers should use common sense and be aware of weather and water conditions. Swimming should stop if people feel or look very cold. People who have seizures that are well controlled need not avoid swimming, but they should be careful near water, whether boating, showering, or bathing.

To decrease the risk of drowning, people should not swim alone and should swim only in areas patrolled by lifeguards. Ocean swimmers should learn to escape rip currents (strong currents that pull away from the shore) by swimming parallel to the beach rather than by swimming toward the beach. People who practice dangerous underwater breath-holding behaviors (DUBBs) should be supervised and should know the dangers of this activity. People do not need to wait an hour after eating to return to swimming. There is no substantial evidence to support the myth that cramps cause drowning when swimming too soon after eating.

Did You Know...

- People do not need to wait an hour after eating to return to swimming.

Other water safety measures

Wearing life jackets approved by a reputable water-safety agency when in boats is encouraged for everyone and is required for nonswimmers and for small children. Spinal injuries can be prevented by not diving into shallow water.

Community swimming areas need to be supervised by lifeguards trained in water safety, [resuscitation](#), and rescue techniques. Life preservers, life jackets, and a shepherd's crook (a long staff with 1 end hooked) should be available close to poolside. Pool areas should have access to automated external defibrillators, equipment for opening airways, and telephones to contact emergency medical services. Comprehensive community prevention programs should do the following:

- Target high-risk groups
- Teach as many adolescents and adults cardiopulmonary resuscitation (CPR) as possible
- Teach children to swim as soon as they are developmentally ready

More Information

The following English-language resources may be useful. Please note that The Manual is not responsible for the content of these resources.

[American Heart Association: CPR & First Aid. Emergency Cardiovascular Care:](#) Information regarding CPR classes, training kits, and community programs. Accessed October 7, 2024.

[Centers for Disease Control and Prevention: Drowning Prevention](#): Includes information about risk factors for and prevention of unintentional drowning. Accessed October 7, 2024.



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